Sparse Autoencoders for LoRA-Adapted CLIP Under Domain Shift

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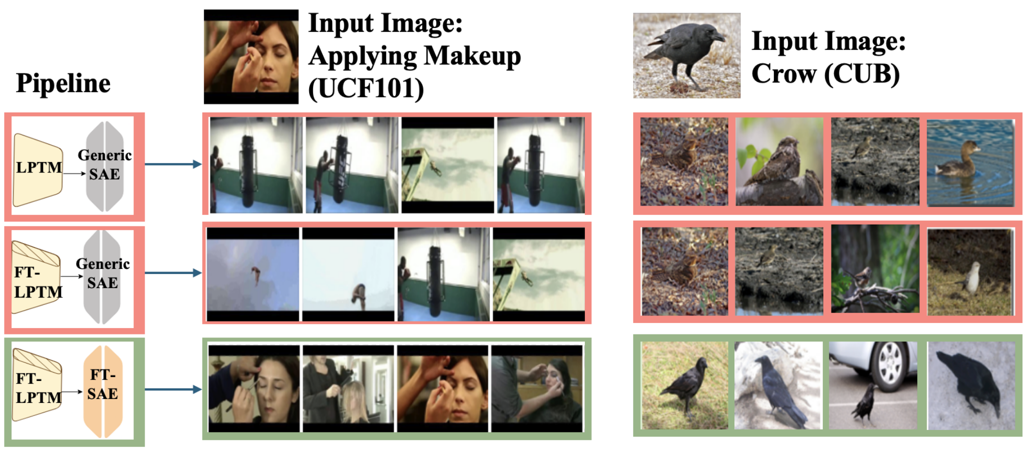

Figure 1: Top-activating images for the per-class top-activating neuron under CLIP and LoRA-adapted CLIP. The FT-SAE trained on adapted activations (bottom row) captures target-domain semantics more faithfully than reusing the Generic SAE (G-SAE) on either base CLIP activations (top row) or adapted CLIP activations (middle row).

Abstract

Sparse autoencoders (SAEs) are increasingly used to interpret CLIP representations, but they are often trained once on base-model activations and then reused after downstream adaptation. We study whether such generic SAEs remain faithful interpreters of LoRA-adapted CLIP under domain shift. Comparing three configurations–base CLIP with a generic SAE, LoRA-adapted CLIP with the same generic SAE, and LoRA-adapted CLIP with a domain-matched SAE–we find that generic SAE reuse becomes unreliable as the target domain moves away from ImageNet-like data. The degradation appears in downstream accuracy after SAE reconstruction, reconstruction error, sparsity, latent usage, and class selectivity. Retraining the SAE on adapted CLIP activations recovers much of this lost fidelity. We further introduce a Domain-Aware Monosemanticity Score (DAMS), which penalizes broad, non-discriminative feature firing that standard top-activation monosemanticity scores can overestimate under shift.

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1 Introduction

CLIP [Radford et al., 2021] is routinely adapted to downstream visual domains where zero-shot performance is insufficient. Parameter-efficient methods such as LoRA [Hu et al., 2022] make this adaptation cheap by adding low-rank updates to selected projection matrices, but the same updates may also move the internal activation distribution away from the one on which existing interpretability tools were trained.

Sparse autoencoders (SAEs) decompose dense CLIP activations into sparse latent features and are increasingly reused as fixed dictionaries for feature-level interpretation [Cunningham et al., 2024, Rajamanoharan et al., 2024, Gandelsman et al., 2024, Lim et al., 2024, Pach et al., 2025]. This reuse assumes that an SAE trained on base CLIP activations remains a faithful basis after adaptation. Under domain shift, however, the adapted model may use directions that the generic dictionary reconstructs poorly or represents with diffuse, non-selective features.

We focus on CLIP ViT-B/16 and LoRA adaptation. This restricted setting lets us isolate a practical question: when a CLIP model is adapted to a target domain, can a generic SAE trained on base CLIP activations still be reused as an interpreter? We evaluate this question across datasets ordered by their MMD distance from ImageNet, comparing generic and domain-matched SAEs using downstream accuracy, reconstruction fidelity, sparsity, latent usage, and target-domain selectivity.

Our contributions are:

1. We test whether a generic SAE trained on base CLIP remains faithful after LoRA adaptation.
2. We show that the generic SAE loses reconstruction fidelity, sparsity efficiency, and downstream utility as domain shift increases.
3. We introduce DAMS, a coverage-aware metric for target-domain monosemanticity.

2 Related Work

CLIP adaptation. CLIP transfers well across many recognition tasks, but specialized domains such as textures, satellite imagery, and medical images often require adaptation [Radford et al., 2021, Hendrycks and Dietterich, 2019]. Prompt learning and lightweight adapters improve downstream accuracy while limiting the number of trainable parameters [Zhou et al., 2022b,a, Gao et al., 2024, Jia et al., 2022]. LoRA [Hu et al., 2022] is especially relevant for our question because it changes internal weight matrices directly, so the adapted activation space need not remain aligned with a dictionary trained on the base model.

SAEs for CLIP interpretability. Sparse autoencoders decompose neural activations into overcomplete sparse features, offering a feature-level alternative to output attribution or architecture-level decomposition [Elhage et al., 2022, Bricken et al., 2023, Cunningham et al., 2024, Rajamanoharan et al., 2024]. Vision and vision–language work uses SAE features as concept bottlenecks, intervention targets, or interpretable CLIP directions [Rao et al., 2024, Stevens et al., 2025, Pach et al., 2025, Gandelsman et al., 2024]. Patch-level SAEs are particularly useful for CLIP because they can expose local texture, object-part, and acquisition cues [Lim et al., 2024].

Gap. Prior SAE analyses commonly train a dictionary once on base activations and reuse it as a fixed interpreter. This is convenient, but it leaves open whether the same dictionary remains faithful after LoRA adaptation to a shifted target domain. We study that reuse assumption directly for CLIP by comparing a generic SAE against an SAE retrained on adapted CLIP activations.

3 Studying SAEs under Domain Shift

We study a narrow, practical reuse question: after CLIP is adapted with LoRA, can an SAE trained on base CLIP activations still be trusted as the interpreter of the adapted model? Let L be the base CLIP visual encoder and $\hat{L}$ its LoRA-adapted version. A *Generic SAE* (G-SAE), $\bar{S}_L$, is trained on base CLIP activations from an ImageNet-like corpus. A *domain-matched SAE* (FT-SAE), $S_{\hat{L}}$, is trained on activations from $\hat{L}$ on the target dataset.

Configurations. We compare three SAE-mediated configurations:

1. **ZS+G-SAE**: $\bar{S}_L(L(I))$, base CLIP reconstructed through the generic SAE;

2. **FT+G-SAE**: $\bar{S}_L(\hat{L}(I))$, LoRA-adapted CLIP reconstructed through the same generic SAE;
3. **FT+FT-SAE**: $S_{\hat{L}}(\hat{L}(I))$, LoRA-adapted CLIP reconstructed through an SAE trained on adapted activations.

The main comparison is FT+G-SAE vs. FT+FT-SAE: if the generic dictionary remains faithful, retraining the SAE should provide little benefit.

3.1 Experimental Setup

Backbone and adaptation. We use CLIP ViT-B/16 [Radford et al., 2021] and adapt it with LoRA applied to the query and value projections. This isolates weight-space adaptation while keeping the study focused on a widely used VLM backbone.

SAEs. The main paper reports Gated SAE results [Rajamanoharan et al., 2024]. Additional SAE variants and broader checks are deferred to the appendix.

Datasets. The main paper uses seven datasets spanning increasing MMD distance from ImageNet: Caltech101 and Oxford Pets as near-domain benchmarks; DTD and UCF101 as mid-shift texture/action benchmarks; and EuroSAT, ChestMNIST, and PathMNIST as far-domain satellite and medical benchmarks [Fei-Fei et al., 2004, Parkhi et al., 2012, Cimpoi et al., 2013, Soomro et al., 2012, Helber et al., 2019, Yang et al., 2023]. The full 15-dataset CLIP table is in Appendix 3.

Metrics. We evaluate downstream classification accuracy after replacing activations with their SAE reconstructions, reconstruction fidelity (NMSE and cosine similarity), sparsity (L_0), dead-neuron fraction, and target-domain monosemanticity. MMD is used to order datasets by distance from ImageNet and to analyze the trend, not as a separate headline hypothesis.

3.2 Hypotheses

1. **H1:** Generic SAE reuse loses downstream utility after CLIP adaptation, especially under larger domain shift.
2. **H2:** Domain-matched SAEs provide a better reconstruction-sparsity tradeoff for adapted CLIP activations.
3. **H3:** Standard monosemanticity can overestimate interpretability under shift; DAMS better reflects target-domain selectivity.

3.3 Results

H1: FT-SAEs recover accuracy that generic SAEs lose. Table 1 shows that reusing the generic SAE on LoRA-adapted CLIP consistently reduces downstream accuracy relative to the adapted model. Retraining the SAE on adapted activations recovers much of that loss: 79–83% recovery on the near and mid-shift benchmarks, 92% on EuroSAT, and 73% on ChestMNIST. PathMNIST is the hardest case: FT-SAE still improves over G-SAE by 3.5 points, but recovers only 55% of the lost adapted-model accuracy, suggesting a larger mismatch between the adapted medical representation and the base CLIP dictionary.

H2: FT-SAEs better match adapted activations. Accuracy alone could reflect a task-specific coincidence. Figure 2 tests the stronger requirement that the FT-SAE reconstructs the adapted activation distribution more efficiently. The largest reconstruction improvement appears on DTD,

Table 1: **H1: Accuracy after SAE reconstruction for LoRA-adapted CLIP ViT-B/16.** Recovery is $(\text{FT+FT-SAE} - \text{FT+G-SAE})/(\text{FT} - \text{FT+G-SAE})$. The domain-matched SAE recovers most of the accuracy lost when the generic SAE is reused on adapted activations.

Dataset	Tier	MMD	FT	FT+G-SAE	FT+FT-SAE	Recovery
Caltech101	Near	0.028	96.1	93.2	**95.6**	83%
Pets	Near	0.029	93.5	89.7	**92.7**	79%
DTD	Mid	0.090	72.0	66.8	**71.1**	83%
UCF101	Mid	0.126	77.6	71.3	**75.9**	73%
EuroSAT	Far	0.291	90.8	85.9	**90.4**	92%
ChestMNIST	Far	0.410	68.2	63.4	**66.9**	73%
PathMNIST	Far	0.423	90.5	84.1	**87.6**	55%

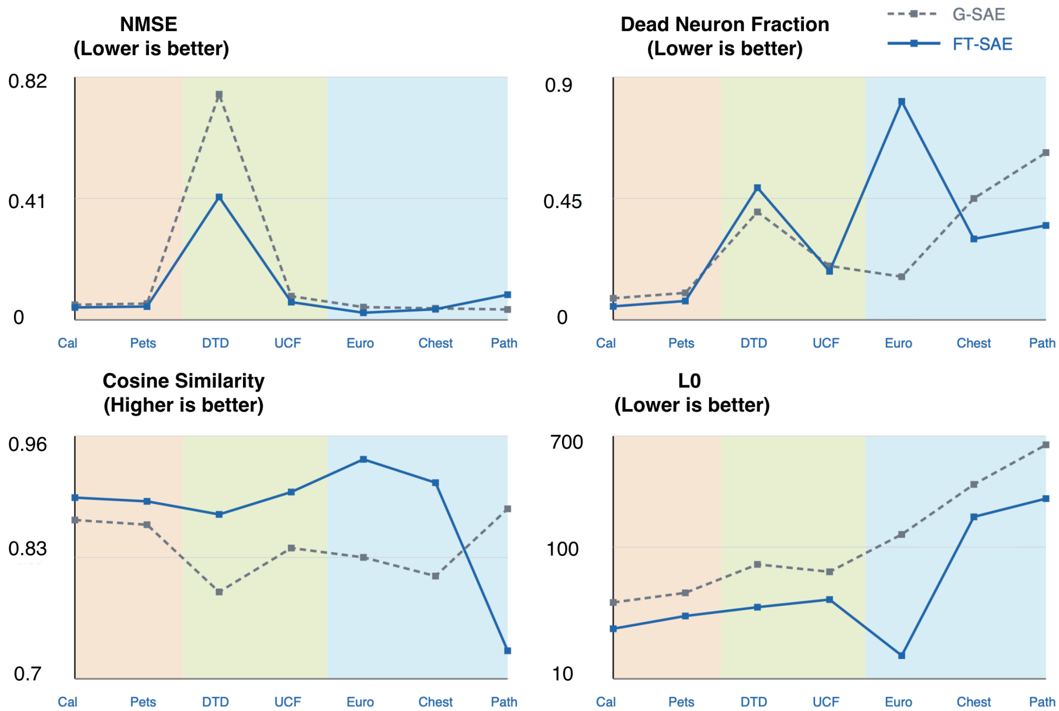

Figure 2: **H2: CLIP-only SAE diagnostics.** Across the main seven datasets, the FT-SAE generally improves reconstruction and sparsity for LoRA-adapted CLIP: NMSE drops on six of seven datasets, cosine similarity rises except on PathMNIST, and L_0 decreases sharply on shifted domains. EuroSAT has a high dead-neuron fraction for the FT-SAE, but also the strongest L_0 reduction, indicating a compact satellite-specific dictionary rather than simple failure.

where NMSE drops from 0.762 to 0.415. On EuroSAT, L_0 falls from 125 to 15 while cosine similarity rises from 0.830 to 0.935, showing that the FT-SAE represents the adapted satellite features with far fewer active latents. The main anomaly is PathMNIST: cosine similarity drops, even though downstream accuracy improves, suggesting that useful medical-domain directions can become less aligned with the base CLIP geometry.

H3: DAMS penalizes broad feature firing under shift. Standard monosemanticity (MS) measures coherence among a latent's top-activating images, but under domain shift a broad catch-all latent can still produce a coherent set while being weakly discriminative for the target task. We therefore use a Domain-Aware Monosemanticity Score:

$$\mathrm{DAMS} = \mathrm{EC}\left[\alpha \cdot \mathrm{CSS}_{\mathrm{norm}} + (1-\alpha) \cdot \mathrm{FSS}\right], \tag{1}$$

where EC is effective coverage, $\mathrm{CSS}_{\mathrm{norm}}$ measures class separability, and FSS measures per-feature class specificity. Full definitions are in Appendix A.2.

Table 2: **H3: MS vs. DAMS across CLIP datasets ordered by MMD. MS**: activation-weighted semantic coherence [Pach et al., 2025]. **DAMS**: Domain-Aware Monosemanticity Score (ours).

	MS		DAMS		
Dataset	G-SAE	FT-SAE	G-SAE	FT-SAE	Δ
Caltech101	0.657	0.610	0.620	0.563	-0.057
DTD	0.520	0.540	0.300	**0.585**	+0.285
UCF101	0.687	0.690	0.678	**0.818**	+0.140
FGVC	0.743	0.760	0.280	**0.500**	+0.220
EuroSAT	0.750	0.756	0.676	0.629	-0.047
PathMNIST	0.840	0.880	0.682	**0.980**	+0.298

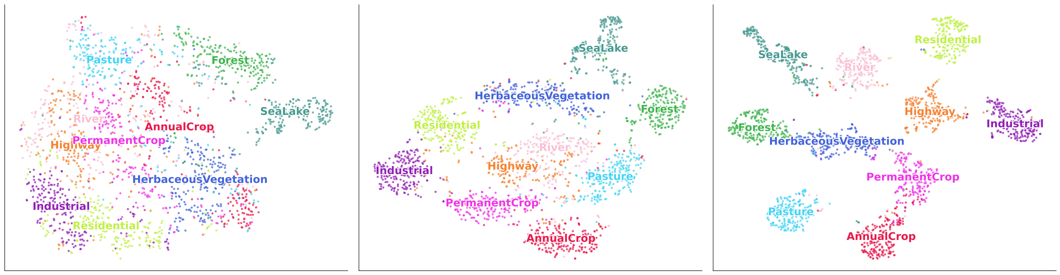

Figure 3: **t-SNE visualisations of SAE latent activations: G-SAE vs FT-SAE for datasets EUROSAT(top) and PathMNIST(Bottom)** Each panel shows patch-level SAE activations projected to 2D via t-SNE, coloured by class label. Domain-matched SAEs produce more compact, class-separable clusters, especially on far-domain benchmarks, consistent with the lower label entropy and higher downstream accuracy in Table **??** and Table 1.

Table 2 shows why DAMS is useful. MS remains high on far-domain datasets, including EuroSAT and PathMNIST, but DAMS separates generic and domain-matched dictionaries more clearly when the target labels require different features. On DTD and PathMNIST, DAMS increases by 0.285 and 0.298 after SAE retraining, respectively, even though the corresponding MS changes are much smaller.

4 Analysis, Limitations and Conclusion

The CLIP-LoRA results support a simple practical rule: the farther the target domain moves from ImageNet-like data, the less safe it is to reuse a generic SAE without checking reconstruction and selectivity. Near-domain datasets still show a measurable gap, but the main failure mode becomes clearer on specialized and far-domain data: the adapted model uses directions that the base dictionary reconstructs with more active latents, poorer coverage, or weaker class specificity.

Why generic SAEs fail. A generic SAE is not only a decoder for a model architecture; it is a sparse dictionary for the activation distribution on which it was trained. LoRA adaptation changes the activations being explained while the generic dictionary stays fixed. When the target domain is close to ImageNet, many features remain reusable. When the domain shifts to satellite or medical images, the adapted model can encode new combinations of texture, acquisition, and class-specific cues that are poorly represented by the base dictionary.

When to retrain. For CLIP-LoRA, retraining is most justified when FT+G-SAE accuracy drops relative to FT, NMSE rises on adapted activations, L_0 becomes diffuse, or DAMS indicates weak target-domain selectivity despite high MS. MMD is a useful first warning signal, but it should be paired with these SAE diagnostics rather than treated as a decision rule by itself.

Limitations. The main paper intentionally studies one backbone and one adaptation method: CLIP ViT-B/16 with LoRA. This makes the claim narrower but cleaner. The appendix includes broader checks, but the main conclusion should be read as a focused CLIP-LoRA result. We also do not perform a full causal circuit analysis of how LoRA changes individual features, and DAMS uses target labels, so it is most directly applicable when labeled validation data is available.

Conclusion. Generic SAEs trained on base CLIP activations should not automatically be reused after LoRA adaptation under domain shift. In our CLIP study, a domain-matched SAE restores reconstruction, sparsity, downstream utility, and target-domain selectivity across the shifted benchmarks. For practitioners, the safe workflow is to treat SAE reuse as a testable assumption and retrain the SAE when adapted activations drift from the base distribution.

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A Technical appendices and supplementary material

A.1 SAE Variant Formulations

All SAE variants share a linear encoder, a nonlinearity $\sigma(\cdot)$, and a linear decoder trained to reconstruct hooked ViT token activations. Let $z \in \mathbb{R}^d$ be a hooked token vector (CLS or patch) at layer ℓ, and let $d_{\mathrm{SAE}} \gg d$ be the overcomplete latent dimension. The general form is

$$\mathrm{SAE}(z) \;=\; W_D^\top\, h(z) + b_D, \qquad h(z) \;=\; \sigma\big(W_E^\top (z - b_D) + b_E\big) \in \mathbb{R}^{d_{\mathrm{SAE}}}_{\geq 0}, \tag{2}$$

where $W_E \in \mathbb{R}^{d \times d_{\mathrm{SAE}}}$, $W_D \in \mathbb{R}^{d_{\mathrm{SAE}} \times d}$, and b_E, b_D are bias terms.

ReLU SAE [Cunningham et al., 2024]. Uses $\sigma = \mathrm{ReLU}$ with an ℓ_1 penalty:

$$\mathcal{L}_{\mathrm{ReLU}} \;=\; \|\mathrm{SAE}(z) - z\|_2^2 \;+\; \lambda_{\ell_1}\, \|h(z)\|_1. \tag{3}$$

Top-k SAE [Gao et al., 2025]. Retains only the k largest pre-activations:

$$h(z) \;=\; \mathrm{TopK}_k\big(W_E^\top (z - b_D) + b_E\big)\,, \qquad \mathcal{L}_{\mathrm{TopK}} \;=\; \|\mathrm{SAE}(z) - z\|_2^2. \tag{4}$$

Gated SAE [Rajamanoharan et al., 2024]. Decouples gating from magnitude using $\pi(z) = W_E^\top (z - b_D)$:

$$h_{\mathrm{gate}}(z) \;=\; \mathbb{1}(\pi(z) + b_{\mathrm{gate}} > 0) \odot \mathrm{ReLU}(\exp(r) \odot \pi(z) + b_{\mathrm{mag}})\,, \tag{5}$$

where $r \in \mathbb{R}^{d_{\mathrm{SAE}}}$ is a learned rescaling. The loss is:

$$\mathcal{L}_{\mathrm{Gated}} \;=\; \|\mathrm{SAE}(z) - z\|_2^2 \;+\; \lambda_{\ell_1}\, \|\mathrm{ReLU}(\pi(z) + b_{\mathrm{gate}})\|_1 \;+\; \|\hat{z}_{\mathrm{gate}}(z) - z\|_2^2, \tag{6}$$

where $\hat{z}_{\mathrm{gate}}$ is the reconstruction using only the gate pathway.

Matryoshka SAE [Bussmann et al., 2025, Zaremba et al., 2025]. Enforces a nested hierarchy via prefix-truncated reconstructions. Given prefix sizes $m_1 < m_2 < \cdots < m_M = d_{\mathrm{SAE}}$, let $h_{1:m}(z)$ retain only the first m latent entries:

$$\mathcal{L}_{\mathrm{Mat}} \;=\; \sum_{k=1}^{M} \alpha_k \left[\left\| W_D^\top h_{1:m_k}(z) + b_D - z \right\|_2^2 + \lambda_{\ell_1}\, \|h_{1:m_k}(z)\|_1 \right]. \tag{7}$$

PatchSAE [Lim et al., 2024]. Uses the ReLU objective (Eq. (3)) trained on patch tokens. For neuron s on image i, patch-level and image-level activation counts are:

$$a_{ij}[s] \;=\; \mathbb{1}(h_{ij}[s] > \tau)\,, \qquad a_i[s] \;=\; \sum_{j=1}^{N} a_{ij}[s]. \tag{8}$$

A.2 Metric Definitions

$$L_0(z) = \|z\|_0 = \sum_{j=1}^{m} \mathbf{1}\{z_j \neq 0\}.$$

For a sparse autoencoder (SAE), if $z \in \mathbb{R}^m$ is the latent activation vector, then $L_0(z)$ counts how many latent features are active.

A dataset-level version is given by

$$L_0 = \frac{1}{N} \sum_{i=1}^{N} \|z_i\|_0 = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{m} \mathbf{1}\{z_{ij} \neq 0\}, \tag{9}$$

where z_i is the SAE latent activation for input x_i. Lower L_0 indicates sparser activations.

Reconstruction fidelity.

$$E_{\text{rec}} \;=\; \mathbb{E}\big[\|\text{SAE}(z)-z\|_2^2\big]\,. \tag{10}$$

Activated frequency.

$$\text{Freq}[s] \;=\; \frac{|\{i\in\mathcal{I}:h_i[s]>0\}|}{|\mathcal{I}|}. \tag{11}$$

Mean activation.

$$\text{MeanAct}[s] \;=\; \frac{1}{|\{i\in\mathcal{I}:h_i[s]>0\}|}\sum_{\substack{i\in\mathcal{I}\\ h_i[s]>0}} h_i[s]. \tag{12}$$

Label entropy. For latent s with reference set R_s (top-k images by image-level activation), let $\text{sum}_c = \sum_{i\in R_s:y_i=c} h_i[s]$ and $p_c = \text{sum}_c/\sum_{c'}\text{sum}_{c'}$:

$$\text{Ent}[s] \;=\; -\sum_{c\in\mathcal{C}} p_c\log p_c. \tag{13}$$

Monosemanticity. Let x_i be the CLIP embedding of image i and $z_i^h \geq 0$ the activation of latent h on i:

$$\text{MS}_h \;=\; \frac{\sum_{i<j} z_i^h z_j^h \cos(x_i,x_j)}{\sum_{i<j} z_i^h z_j^h}. \tag{14}$$

Full DAMS definition. The main paper uses the compact form of DAMS. In full, we compute

$$\text{DAMS} = \text{EC}^{\rho}\left[\alpha\,\text{CSS}_{\text{norm}} + \beta\,\text{FSS} + \gamma\,\text{DAS}\right], \qquad \alpha{=}(1{-}\gamma)\phi, \quad \beta{=}(1{-}\gamma)(1{-}\phi). \tag{15}$$

Effective Coverage penalizes dead and indiscriminate neurons:

$$\text{EC} = (1-f_{\text{dead}})(1-\bar{d}), \quad f_{\text{dead}} = \frac{|\{j:\mathbf{z}_j=\mathbf{0}\}|}{L}, \quad \bar{d} = \frac{1}{L_{\text{alive}}}\sum_{j\in\mathcal{A}}\frac{|\{i:z_{ij}>0\}|}{N}. \tag{16}$$

Class separability uses a Fisher ratio with saturation normalization:

$$\text{CSS}_{\text{norm}} = \frac{\text{CSS}_{\text{raw}}}{\text{CSS}_{\text{raw}}+\kappa}, \qquad \text{CSS}_{\text{raw}} = \frac{1}{C}\,\text{tr}\big((\mathbf{S}_W{+}\epsilon\mathbf{I})^{-1}\mathbf{S}_B\big)\,. \tag{17}$$

Feature specificity measures entropy-sharpened class purity per alive latent:

$$\text{FSS} = \frac{1}{L_{\text{alive}}}\sum_{j\in\mathcal{A}}(1{-}H_{\text{norm}}(j))^s\,\max_c\,p(c\,|\,j), \qquad p(c\,|\,j) = \frac{\sum_{i\in\mathcal{I}_c} z_{ij}}{\sum_i z_{ij}}. \tag{18}$$

Domain alignment is optional in the compact main-paper metric and is implemented as centered kernel alignment between class-pooled SAE activations and the ideal label kernel:

$$\text{DAS} = \text{CKA}(\mathbf{K}_{\text{SAE}},\,\mathbf{K}_{\text{label}}), \qquad [\mathbf{K}_{\text{label}}]_{cc'} = \mathbf{1}[c{=}c']. \tag{19}$$

A.3 Full CLIP and ablation tables

A.4 SAE statistics and architecture ablations

Table 3: **Full CLIP accuracy table.** Classification accuracy (%) for CLIP ViT-B/16 across the full dataset sweep. Columns are ordered by MMD from ImageNet.

Configuration	Caltech101	Pets	ImageNet-A	SUN397	ImageNet-Sk.	Corruption	Food101	DTD	UCF101	StanfordCars	Flowers102	FGVC	EuroSAT	ChestMNIST	PathMNIST
ZS	93.3	84.9	47.1	62.3	52.4	38.6	78.2	45.2	64.5	58.1	72.4	23.7	45.1	21.5	17.9
FT	96.1	93.5	72.8	76.1	80.1	64.3	90.4	72.0	77.6	79.3	88.2	46.4	90.8	68.2	90.5
ZS+G-SAE	92.1	82.5	44.8	60.2	50.6	36.9	75.1	45.9	61.9	55.8	70.2	21.1	59.6	20.1	18.2
FT+G-SAE	93.2	89.7	68.2	70.2	74.9	58.1	85.3	66.8	71.3	73.8	83.1	38.5	85.9	63.4	84.1
FT+FT-SAE	**95.6**	**92.7**	**71.2**	**74.8**	**78.3**	**62.9**	**89.2**	**71.1**	**75.9**	**77.8**	**86.9**	**44.8**	**90.4**	**66.9**	**87.6**
MMD	0.028	0.029	0.050	0.057	0.070	0.079	0.080	0.090	0.126	0.140	0.152	0.230	0.291	0.410	0.423

Table 4: **Appendix ablation: SAE expansion factor and generic training set.** G-SAE is trained either on ImageNet or on CC3M; FT-SAE is trained on LoRA-adapted CLIP activations.

Method	**Exp.**	Caltech101	Oxford Pets	FGVC	DTD	UCF101	EuroSAT	PathMNIST
LoRA-FT	—	96.1	93.5	46.4	72.0	77.6	90.8	90.5
G-SAE (IN)	×32	89.8	85.4	33.2	62.1	66.8	81.3	78.5
G-SAE (IN)	×64	92.4	88.6	37.1	65.8	70.4	85.0	82.9
G-SAE (IN)	×128	93.1	89.5	38.4	66.9	71.6	86.2	83.8
G-SAE (CC3M)	×64	47.8	28.3	5.8	35.7	43.3	49.8	31.0
FT-SAE	×64	**95.7**	**92.4**	**44.5**	**71.8**	**76.2**	**90.7**	**88.7**

Table 5: **H2: Reconstruction quality and feature utilisation.** CosSim ↑, NMSE ↓, Dead-neuron fraction ↓ for G-SAE and FT-SAE across all four backbones. Columns ordered by Maximum Mean Discrepancy from ImageNet (near → far). **Bold**: best result per metric per backbone per dataset.

Encoder	**Metric**	**SAE**	Caltech101	Pets	ImageNet-A	SUN397	ImageNet-Sk.	Corruption	Food101	DTD	UCF101	StanfordCars	Flowers102	FGVC	EuroSAT	ChestMNIST	PathMNIST
CLIP	CosSim↑	G-SAE	0.870	0.865	0.800	0.832	0.805	0.795	0.835	0.793	0.840	0.855	0.860	0.850	0.830	0.810	0.882
		FT-SAE	**0.894**	**0.890**	**0.890**	**0.905**	**0.895**	**0.885**	**0.910**	**0.876**	**0.900**	**0.880**	**0.885**	**0.875**	**0.935**	**0.910**	0.730†
	NMSE↓	G-SAE	0.051	0.055	0.045	0.052	0.044	0.046	0.050	0.762	0.080	0.065	0.060	0.058	0.043	0.039	0.035
		FT-SAE	**0.042**	**0.045**	**0.038**	**0.034**	**0.037**	**0.039**	**0.035**	**0.415**	**0.060**	**0.050**	**0.048**	**0.047**	**0.024**	**0.036**	0.085†
	Dead↓	G-SAE	0.08	0.10	0.40	0.17	0.42	0.39	0.18	0.40	0.20	0.15	0.12	0.11	0.16	0.45	0.62
		FT-SAE	**0.05**	**0.07**	**0.28**	**0.13**	**0.29**	**0.27**	**0.14**	0.49†	**0.18**	**0.10**	**0.08**	**0.09**	0.81†	**0.30**	**0.35**

Table 6: **H1 (additional benchmarks): Classification accuracy (%) on OfficeHome, KITTI, and Cityscapes.** $\bar{S}$: generic/base SAE. $\hat{S}$: domain-matched SAE (LoRASAE). The generic SAE degrades after LoRA adaptation, while the FT-SAE partially recovers performance.

Configuration	OfficeHome	KITTI	Cityscapes
ZS	83.02	32.78	11.2
LoRA-FT	87.25	64.72	47.6
ZS + $\bar{S}$	62.30	21.50	11.2
LoRA-FT + $\bar{S}$	69.45	58.70	6.5
LoRA-FT + $\hat{S}$	**86.90**	**61.67**	**42.01**

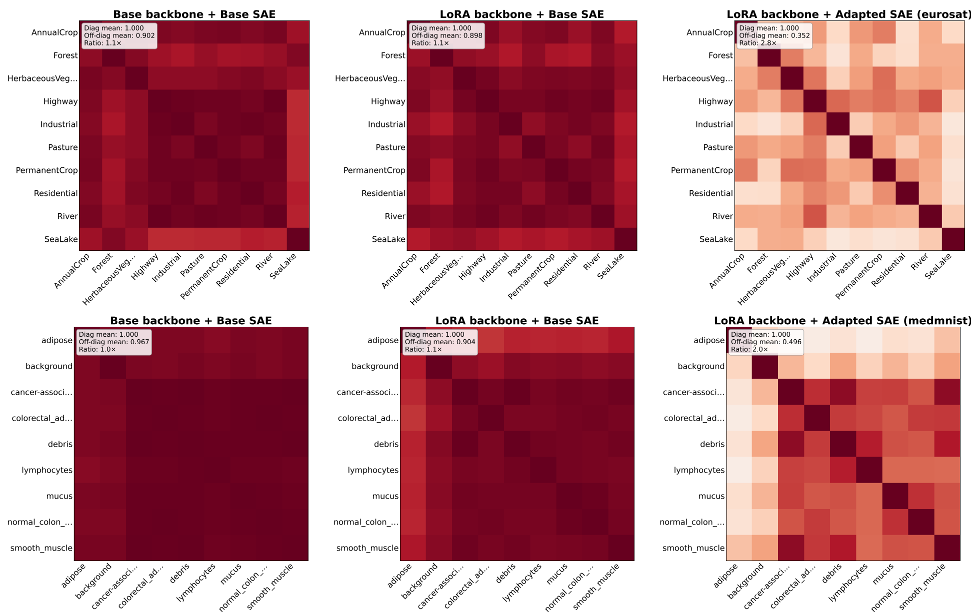

Figure 4: **Class-level cosine-similarity heat maps.** Top row: EuroSAT; bottom row: PathMNIST. Left: ZS+G-SAE; middle: FT+G-SAE; right: FT+FT-SAE. Domain-matched SAEs learn more class-discriminative features.

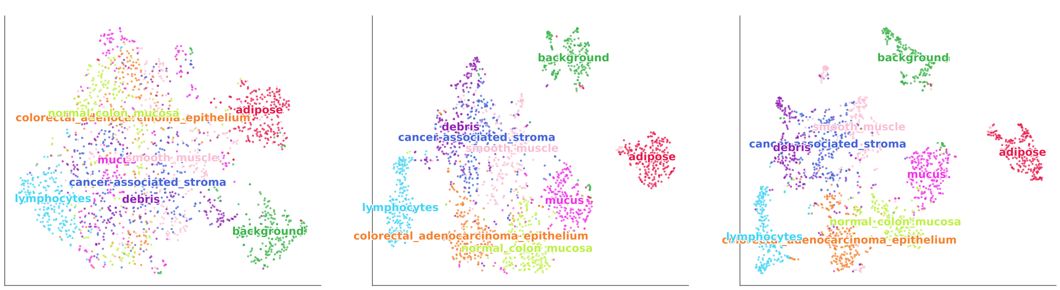

Figure 5: **t-SNE visualizations of SAE latent activations.** Each panel shows patch-level SAE activations projected to 2D and colored by class label. Domain-matched SAEs produce more compact, class-separable clusters on far-domain benchmarks.